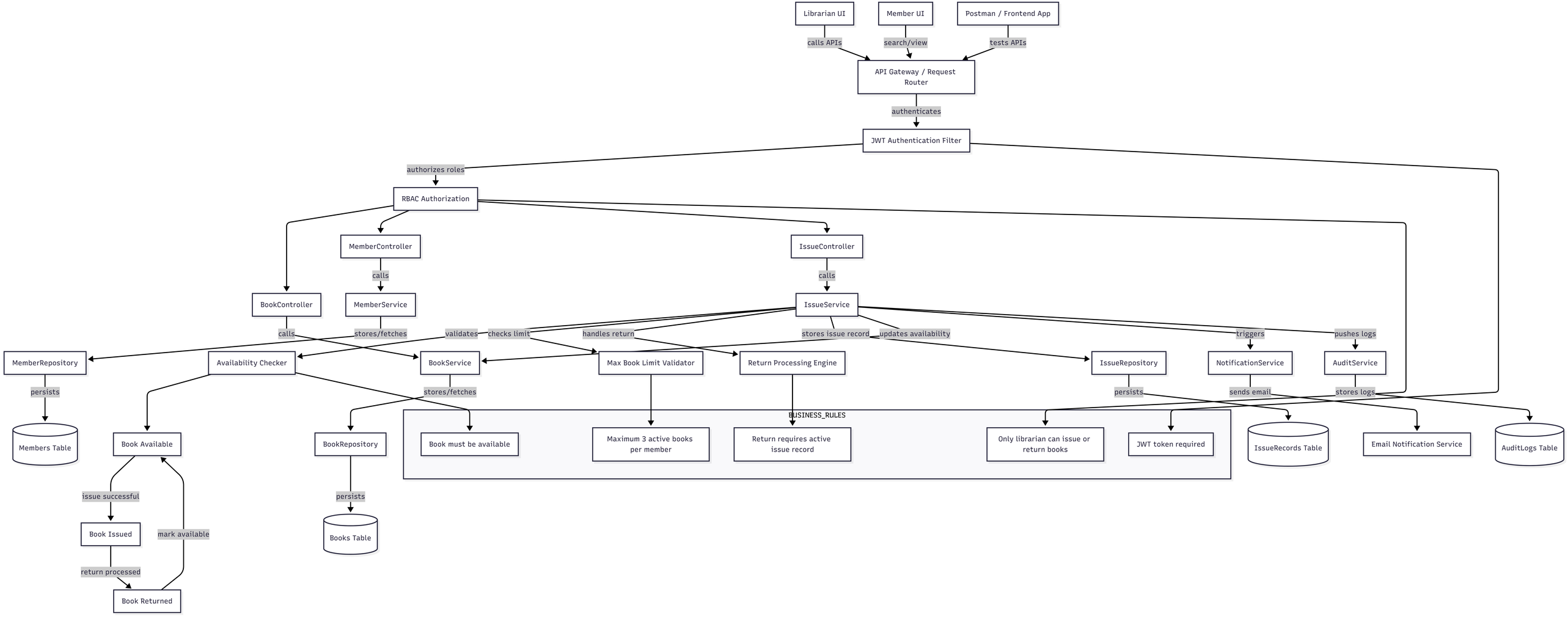
Librarian UI
Member UI
Postman / Frontend App
calls APIs
search/view
tests APIs
API Gateway / Request Router
authenticates
JWT Authentication Filter
authorizes roles
RBAC Authorization
MemberController
IssueController
calls
calls
BookController
MemberService
IssueService
calls
stores/fetches
validates
checks limit
handles return
stores issue record
updates availability
triggers
pushes logs
MemberRepository
Availability Checker
BookService
Max Book Limit Validator
Return Processing Engine
IssueRepository
NotificationService
AuditService
persists
stores/fetches
persists
sends email
stores logs
BUSINESS_RULES
Members Table
Book Available
BookRepository
Book must be available
Maximum 3 active books per member
Return requires active issue record
Only librarian can issue or return books
JWT token required
IssueRecords Table
Email Notification Service
AuditLogs Table
issue successful
persists
Book Issued
mark available
Books Table
return processed
Book Returned